The four-stroke engine

Most car engines are four-stroke piston engines. A piston slides up and down inside a cylinder; a connecting rod links it to the crankshaft, which turns that up-and-down motion into rotation. Two valves at the top of the cylinder, the intake valve and the exhaust valve, let gases in and out. A camshaft, driven by the crankshaft at half its speed, opens each valve at the right moment.

The four strokes

Intake. The piston moves down with the intake valve open, drawing a mixture of air and fuel into the cylinder.

Compression. Both valves close and the piston moves up, squeezing the mixture into a small space. The compression ratio is the cylinder volume at the bottom of the stroke divided by the volume at the top; petrol engines use about 10:1.

Power. The spark plug ignites the mixture. The burning gas expands and pushes the piston down hard; this is the only stroke that delivers work.

Exhaust. The exhaust valve opens and the piston moves up, pushing the burnt gas out.

Why a flywheel

Only one stroke in four pushes the crankshaft. A heavy flywheel stores energy from the power stroke and carries the engine through the other three, so the crankshaft keeps turning smoothly.

Key terms

Top dead centre (TDC): the piston's highest point. Bottom dead centre (BDC): its lowest point. Stroke: the distance between them. Displacement: the volume the piston sweeps in one stroke, times the number of cylinders.